Brno University of Technology
Faculty of Electrical Engineering and Communication

Course:
Machine Learning Fundamentals (MPA-MLF)

Classification of Room Occupancy

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1 Introduction

Room occupancy estimation is a key problem in smart building management, energy efficiency, and security systems. Traditional approaches based on passive infrared sensors or cameras raise significant privacy concerns. An alternative that preserves anonymity while remaining highly effective relies on millimetre-wave radar signals, specifically at 60 GHz, which are capable of detecting human presence and movement without capturing any visual information.

The objective of this project is to classify the number of persons present in a room by analysing two-dimensional snapshots of radar signals in the delay-Doppler domain. Each snapshot encodes how much of the transmitted signal was reflected, at what delay (linked to distance), and at what Doppler frequency shift (linked to velocity). The task is a four-class classification problem, where the possible labels are: zero persons (machine only), one person, two persons, or three persons.

The approach adopted in this report relies on a Convolutional Neural Network (CNN), which is well-suited for extracting spatial features from two-dimensional images. The model is trained on 9,227 labelled snapshots and evaluated on the Kaggle competition leaderboard. The final model achieved a public leaderboard score of 0.95182 on the Kaggle platform.

The complete source code is available in the project GitHub repository at: https://github.com/KevinLny/MPA-MLF-Project-FIT.

2 Data Analysis and Domain Description

2.1 The Delay-Doppler Domain

Each input sample is a PNG image representing a two-dimensional snapshot of a 60 GHz radar signal in the delay-Doppler domain. The horizontal axis encodes the delay, which is proportional to the distance between the receiver and the target: the further the person is, the greater the delay. The vertical axis encodes the doppler frequency shift, which is proportional to the radial velocity of the target: a faster-moving person generates a higher Doppler shift.

In ideal conditions, each person in the room appears as a distinct bright spot at a unique position in this two-dimensional map. However, real-world records suffer from noise, multipath reflections, missed targets, and interference from static objects (referred to as "machine only" in the zero-person class). Each image has a native resolution of 45×51 pixels with 3 colour channels (RGB), totalling 6,885 values per sample.

2.2 Class Distribution and Dataset Examination

The training set consists of 9,227 labelled images. The class distribution is presented in Table 1. A significant class imbalance is observed: class 1 (one person) accounts for over 41% of the training data, while class 3 (three persons) represents only about 10.6%. This imbalance was taken into account when interpreting the per-class performance metrics.

Table 1: Class distribution in the training dataset.

Class	**Number of Samples**	**Proportion (%)**
0 – Machine only (0 persons)	1,462	15.8%
1 – One person	3,786	41.0%
2 – Two persons	3,002	32.5%
3 – Three persons	977	10.6%
Total	**9,227**	**100%**

3 Data Preprocessing

3.1 Original Image Resolution

Unlike preliminary versions of the pipeline that resized all images to a fixed 64×64 resolution, the final model operates on the original image dimensions of 45×51 pixels. Resizing was initially considered to standardise the input shape and reduce memory usage. However, all images in the dataset already share the same native resolution, making resizing unnecessary. More importantly, downscaling introduces interpolation artefacts that can blur the distinctive bright spots representing persons in the delay-Doppler domain. Preserving the native resolution therefore retains all discriminative spatial information at negligible extra cost.

3.2 Normalisation and Feature Scaling

All pixel values, originally encoded as 8-bit unsigned integers in the range $[0, 255]$, are converted to 32-bit floating point and rescaled to $[0, 1]$ by dividing by 255. This normalisation ensures that all input features have a comparable magnitude, which accelerates gradient descent convergence and improves numerical stability during training.

3.3 Dataset Split Strategy

The full training set of 9,227 images is used to train the model without any held-out validation split during training. Since the Kaggle competition provides an independent test set that serves as the ultimate evaluation benchmark, keeping all labelled data for training maximises the information available to the model. An a posteriori evaluation on the last 20% of training samples (1,846 images, fixed by index order) is performed separately to compute the confusion matrix and per-class metrics, without influencing the training process.

4 Model Architecture

4.1 Justification for a Convolutional Neural Network

A Convolutional Neural Network is the natural choice for this task for several reasons. First, the input data are two-dimensional images where spatial structure is meaningful: the position of each bright spot encodes the physical location and velocity of a person. Convolutional layers are specifically designed to detect local spatial patterns regardless of their exact position in the image, a property known as translation equivariance. Second, compared to a fully connected network applied directly to raw pixels, a CNN drastically reduces the number of trainable parameters through weight sharing, which limits overfitting on a dataset of moderate size. Finally, CNNs have been extensively validated on image classification tasks, providing a strong and well-understood baseline.

4.2 Architecture Overview

The model follows a classic encoder-classifier structure. Two convolutional blocks progressively extract spatial features at increasing levels of abstraction, followed by a fully connected classifier head that maps the learned representation to the four output classes. The complete architecture is illustrated in Figure 1 and summarised in Table 2.

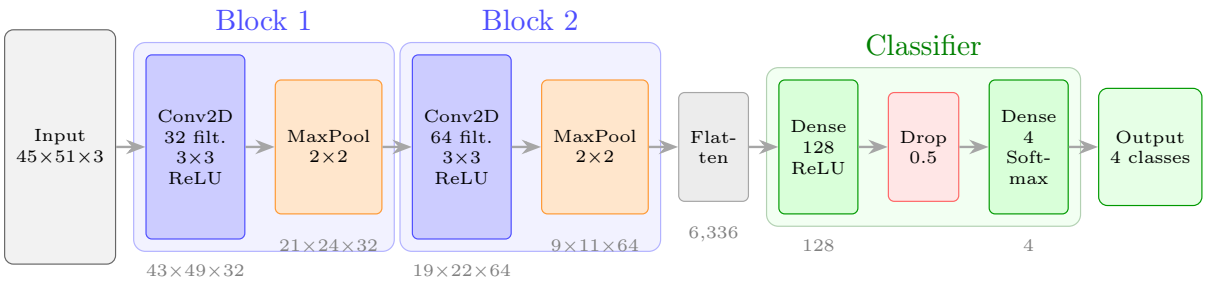

Figure 1: Architecture of the CNN model. Feature map shapes are indicated below each layer.

Table 2: Detailed layer description of the CNN model.

Layer	**Type**	**Output Shape**	**Parameters**
Input	Input	$45 \times 51 \times 3$	0
Conv2D-1	Conv2D (ReLU)	$43 \times 49 \times 32$	896
MaxPooling-1	MaxPool 2×2	$21 \times 24 \times 32$	0
Conv2D-2	Conv2D (ReLU)	$19 \times 22 \times 64$	18,496
MaxPooling-2	MaxPool 2×2	$9 \times 11 \times 64$	0
Flatten	Flatten	6,336	0
Dense-1	Dense (ReLU)	128	811,136
Dropout	Dropout (0.5)	128	0
Dense-2	Dense (Softmax)	4	516
Total			**831,044**

4.3 Layer-by-Layer Description

Convolutional Layers: The two convolutional layers use 3×3 filters with ReLU activation and no padding, so each convolution reduces the spatial dimensions by 2 pixels on each side. The first layer applies 32 filters, learning low-level features such as edges and intensity gradients. The second applies 64 filters, combining these primitive features into higher-level spatial patterns that correspond to the signatures of persons in the delay-Doppler map.

Max Pooling: Each convolutional layer is followed by a 2×2 max-pooling operation, which approximately halves the spatial dimensions while retaining the most prominent activations. This reduces the computational cost of subsequent layers and introduces a degree of translation invariance, making the model more robust to small displacements of the radar spots.

Flatten and Dense Layers: After the two convolutional blocks, the feature maps are flattened into a one-dimensional vector of length 6,336. This vector is passed through a fully connected layer of 128 neurons with ReLU activation, which learns the final non-linear combination of spatial features relevant to counting the number of persons.

Dropout: A Dropout layer with rate 0.5 is applied between the two dense layers. During training, this randomly deactivates 50% of the neurons at each step, preventing co-adaptation and acting as an effective regulariser against overfitting.

Output Layer: The final dense layer contains 4 neurons corresponding to the four occupancy classes. A Softmax activation converts the raw scores into a probability distribution over the classes, and the class with the highest probability is selected as the final prediction.

4.4 Loss Function and Optimiser

The model is trained using categorical cross-entropy as the loss function, which is the standard choice for multi-class classification when labels are one-hot encoded. The Adam optimiser is used with its default learning rate of 10^{-3}, combining the benefits of adaptive per-parameter learning rates (AdaGrad) and momentum (RMSProp) for fast and stable convergence.

5 Experimental Results and Evaluation

5.1 Kaggle Competition Performance

The final model was submitted to the Kaggle competition associated with this project. The model achieved a **public leaderboard score of 0.95182**, reflecting strong generalisation to unseen data.

The private leaderboard score, evaluated on the remaining 50% of test images initially hidden, is **0.96318**.

5.2 Internal Validation: Confusion Matrix

To analyse the per-class behaviour of the classifier, predictions were generated on the last 20% of training images (1,846 samples) and compared against the ground-truth labels. The resulting confusion matrix is shown in Figure 2.

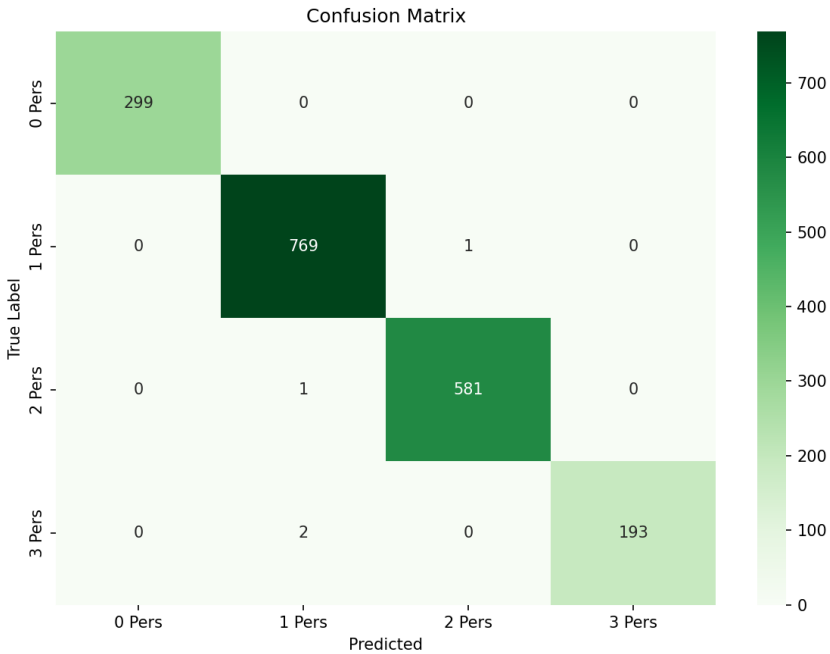

Figure 2: Confusion matrix on the internal validation set (last 20% of training data, 1,846 samples).

The matrix reveals near-perfect classification across all four classes. Out of 1,846 samples, only 4 were misclassified, yielding an internal validation accuracy of **99.78%**. The errors are concentrated at the boundary between adjacent classes: one sample from class 1 was predicted as class 2, one sample from class 2 was predicted as class 1, and two samples from class 3 were predicted as class 1. This pattern is physically interpretable: in noisy delay-Doppler snapshots, the radar signature of one person may be attenuated or masked, causing the classifier to underestimate the occupancy count by one.

5.3 F1-Score and Classification Metrics

Table 3 presents the per-class precision, recall, and F1-score computed from the internal validation set.

Table 3: Classification report on the internal validation set.

Class	Precision	Recall	F1-Score	Support
0 – Machine only	1.00	1.00	1.00	299
1 – One person	1.00	1.00	1.00	770
2 – Two persons	1.00	1.00	1.00	582
3 – Three persons	1.00	0.99	0.99	195
Weighted avg	**1.00**	**1.00**	**1.00**	**1,846**

All classes achieve a precision and recall of 1.00 with the exception of class 3, which records a recall of 0.99 due to the two misclassified three-person samples. This slight deficit is consistent with the lower representation of class 3 in the training set (10.6% of samples), which provides fewer examples for the model to learn from. Nevertheless, the overall performance is excellent across all classes, indicating that the model generalises well and does not exhibit strong bias towards the majority class.

5.4 Learning Curves

Figure 3 shows the evolution of training accuracy and loss over 30 epochs.

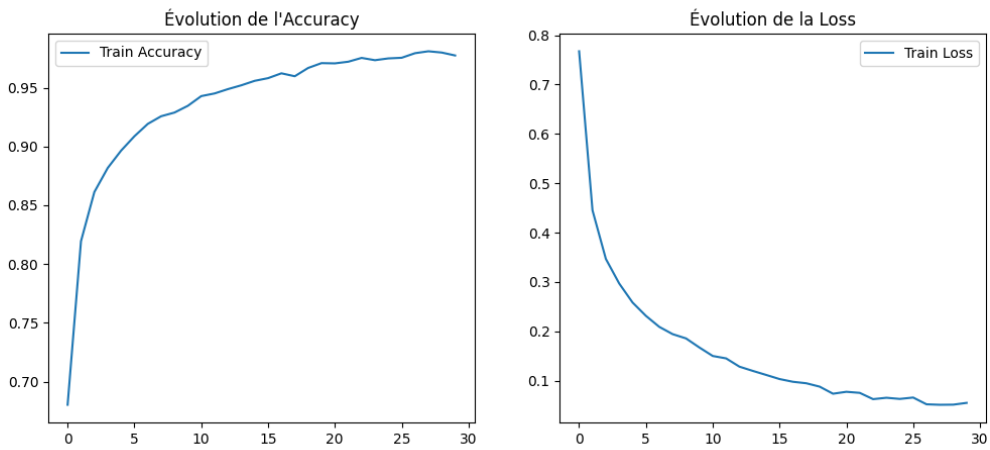

Figure 3: Training accuracy (left) and training loss (right) over 30 epochs.

The accuracy curve rises steeply during the first 10 epochs, reaching approximately 93%, before continuing to improve more gradually to a final value of approximately **97.75%** at epoch 30. Correspondingly, the loss drops sharply from an initial value of 0.77 to below 0.10 within the first 10 epochs, then decreases more slowly to a final value of approximately **0.055**. The smooth, monotonically decreasing loss confirms that the training process was stable and that the optimiser converged correctly. Since no validation loss curve is displayed (the model is trained on the full dataset), the risk of overfitting

is assessed indirectly through the Kaggle submission score and the a posteriori confusion matrix, both of which confirm strong generalisation to unseen data.

6 Optimisation Strategies

The final model was reached through an iterative development process. The main design decisions and their impact on performance are discussed below.

The final model was reached through an iterative development process. The main design decisions and their impact on performance are discussed below:

① **Preserving the Original Image Resolution:**
An initial version of the pipeline resized all images to 64×64 pixels to ensure a uniform input shape, which is a common practice in image classification. However, since all images in the dataset already share the same native resolution of 45×51 pixels, resizing is unnecessary. More importantly, downscaling blurs the distinctive bright spots that represent persons in the delay-Doppler domain. Removing the resize step preserved all discriminative spatial information and simplified the preprocessing pipeline.

② **Training on the Full Dataset:**
Early versions of the script held out 20% of the data as a validation set during training, which reduced the effective training set to 7,381 samples. This split was removed in the final version so that the model could learn from all 9,227 available labelled samples. The Kaggle competition provides an independent test set that serves as the ultimate evaluation benchmark, removing the need for an internal validation split to guide model selection.

③ **Architecture Depth and Filter Count:**
The two-block convolutional architecture with 32 and 64 filters was selected as a good balance between model capacity and training speed on a CPU. Deeper architectures involving three convolutional blocks, batch normalisation layers, and up to 128 filters were tested but did not yield a significant improvement on the Kaggle leaderboard while substantially increasing training time. The simpler architecture converges faster and proved sufficient for the complexity of the task.

④ **Dropout Regularisation:**
A Dropout rate of 0.5 applied after the first dense layer was found to be effective at preventing overfitting. This is particularly important when training on the full dataset, as no held-out validation set is available to detect early overfitting during the training loop.

⑤ **Hyperparameter Selection:**
The model was trained for 30 epochs with a batch size of 32. These values were chosen empirically: the learning curves indicate that the model has not fully converged at epoch 30 (accuracy is still slowly improving), suggesting that additional epochs could yield marginal gains. The batch size of 32 provides a good trade-off between gradient noise (small batches) and memory consumption (large batches).

7 Conclusion

This report presented a Convolutional Neural Network for the classification of room occupancy from 60 GHz radar snapshots in the delay-Doppler domain. The model consists of two convolutional blocks followed by a fully connected classifier, trained with categorical cross-entropy and the Adam optimiser over 30 epochs on 9,227 labelled images at their native resolution of 45×51 pixels.

The final model achieved a public Kaggle leaderboard score of 0.95182, demonstrating strong generalisation to unseen test data. An internal evaluation on the last 20% of training samples yielded a validation accuracy of 99.78% with near-perfect performance across all four occupancy classes (precision and recall of 1.00 for classes 0, 1, and 2; recall of 0.99 for class 3). The small number of remaining errors, concentrated at the boundary between adjacent classes, is consistent with the inherent difficulty of distinguishing between k and $k + 1$ persons when one person's radar signature is weak or overlapping with another.

Key design decisions that contributed to the final performance include preserving the native image resolution, training on the complete labelled dataset, and applying Dropout regularisation to mitigate overfitting. Future improvements could explore data augmentation techniques tailored to the delay-Doppler domain, transfer learning from pretrained feature extractors, or ensembling multiple CNN models to further reduce classification errors on the minority class.